

Introduction to Natural Language Processing

HW 1

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Task introduction

- **Topic** : Multi-label Classification in NLP
- Goal : Apply deep learning model to predict the concern in tweets
 - Implement at least one recurrent neural network model (RNN or LSTM) to predict concerns in tweets (multi-label classification).
 - Use a Transformer-based model (e.g., BERT) for the same task and compare its performance with your RNN/LSTM model.
- You may use any external packages, such as PyTorch, Keras, scikit-learn.
- Pretrained models (e.g., BERT) are allowed, but the total number of model parameters must be **under 1 billion (1B)**.
- Large Language Models (LLMs) are not allowed in HW1.
- Requirement :
 1. Submit a report and your source code to E3
 2. Upload your submission to Kaggle

Dataset

COVID-19 anti-vaccine tweets labelled with various specific anti-vaccine concerns in a multi-label setting.

- HW1_dataset.zip
 - train.json
 - val.json
 - test.json
 - sample_submission.csv



Data Introduction

Each object in the json file contains information about a tweet and its associated labels, including:

- 'ID' : unique identifier of the tweet
- 'tweet' : the content of the tweet
- 'labels' : concerns that have been identified within the tweet with the specific terms causing the concerns and their positions. **Each tweet can have multiple concerns!**

```
{'ID': '1349576125454852097',  
  'tweet': '@4mishee @ScottMorrisonMP can keep his vaccine 🦋 I'll take covid with its " non existent " symptoms over some b  
ig pharma no liability or indemnity lab 🦋 cooked vaccine thank you.',  
  'labels': {'pharma': [{'index': 0,  
    'start': 19,  
    'end': 25,  
    'terms': 'big pharma no liability or indemnity'}],  
    'unnecessary': [{'index': 0,  
      'start': 13,  
      'end': 17,  
      'terms': 'non existent " symptoms"}]}}
```

Kaggle Submission Format

Your model is expected to classify the concern of the tweets in 'test.json' file, and then upload your model's predictions to Kaggle. the submission format should be :

- A 1977*13 .csv file, first row for column name and the last 1976 rows for your result ('1' indicates that the tweet has the concern, '0' signifies that it does not)
- First row must match the one shown in the sample_submission.csv, make sure the order is correct !

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	index	ineffective	unnecessary	pharma	rushed	side-effect	mandatory	country	ingredients	political	none	conspiracy	religious	
2	0	0	0	0	0	0	0	0	0	0	0	0	0	
3	1	0	0	0	0	0	0	0	0	0	0	0	0	
4	2	0	0	0	0	0	0	0	0	0	0	0	0	

Kaggle Submission (70%)

- [Kaggle link](#)
- Use your [@nycu.edu.tw](#) email address to join the competition and change your team name with your student ID, we use this information for grading.
- There is a simple baseline and a strong baseline, beat them to get higher score.

Public Private

This leaderboard is calculated with approximately 30% of the test data. The final results will be based on the other 70%, so the final standings may be different.

#	Team	Members	Score	Entries	Last
	Strong Baseline		0.6246	1	2h
	Simple Baseline		0.4577	1	7m

Kaggle Submission (70%)

- Public leaderboard is calculated with approximately 30% of the test data, private leaderboard is calculated with the other 70%, so the final standings may be different.
You can only view your private leaderboard score after the competition has ended.
- The scoring metric is **macro-F1**, higher score means better performance.
- You can submit **at most 5 times each day** and choose 3 of the submissions to be considered for the private leaderboard, or will otherwise default to the best public scoring submissions.

Change your team name

Please remember to change the team name to <student ID>, or there will be a deduction of 5 points for HW 1.

2026 INLP HW1

Multi-label Classification in NLP



Overview Data Discussion Leaderboard Rules **Team** Submissions Settings

Your Team

Everyone that competes in a Competition does so as a team - even if you're competing by yourself. [Learn more.](#)

General

TEAM NAME

yourStudentID

This name will appear on your team's leaderboard position.

Report Submission

Please answer the following 3 questions, provide your thinking process as detailed as possible:

1. (5%) Describe how you build your model ?

(5%) How did you do to preprocess your data from dataset ? The distribution of the concern is imbalanced, what did you do to improve the macro F1 score on those concern which are in small scale ?

2. When implementing your model:

(5%) Have you tried pretrained word embedding ? (e.g. Glove or Word2vec) What was their influence on the result after using them ?

(5%) Have you tried attention on your model ? What was its influence on the result? When your model predict the concern, what was it focusing on ? Do some case studies.

Report Submission

3. (10%) Compare the differences in performance between your RNN/LSTM model and the Transformer-based model. Which approach performs better? Explain the possible reasons for the difference.

Please answer the questions in detail to get the full point of each question.

Grading policy

- Kaggle (70%)
 - 30% based on the public leaderboard score and 70% based on the private leaderboard score
 - Basic score :
 - Over strong baseline : 55
 - Over simple bassline : 40
 - Under simple baseline : 25
 - Ranking score:
 $15 - (15/N) * (\text{ranking} - 1)$, N=numbers of people in the interval
- Report (30%)
 - 10% for each quesiton

You will receive 0 points if you do not submit the source code.

E3 Submission

Submit your source code and report to E3 before 4/7(Tue) 23:59, **no late submissions will be accepted !**

Format:

- HW1_<student ID>.zip
 - source code : HW1_<student ID>.py or HW1_<student ID>.ipynb
 - report : HW1_<student ID>.pdf

If you have any question about HW 1, please feel free to contact with TA: You-Mei Huang through email ymh636.cs13@nycu.edu.tw

Have Fun !

